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COURSE NAME: Database Management Systems

COURSE CODE: CSA0503

CO1: Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships.

Activity Tool : Short Answer Text(High)

Assessment Tool Weightage: 30%

Question 2: Course Management System (ER Diagram)

Consider a university database for the scheduling of classrooms for final exams. This database could be modeled as the single entity set Exam with attributes course-name, section-number, room-number, and time. Alternatively, one or more additional entity sets could be defined, along with relationship sets to replace some of the attributes of the Exam entity set, as:

- Course with attributes Name, Department, Course_Number
- Section with attributes Section_Number, Enrollment, and dependent as a Weak Entity on Course
- Room with attributes Room_Number, Capacity, Building

Show an E-R diagram illustrating the use of all three additional entity sets listed.

Explanation

Instead of storing all information in a single Exam entity, the database is divided into multiple entities to avoid redundancy and improve data organization.

The main entities are:

1. Course
2. Section (Weak Entity)
3. Room
4. Exam

Step 1: Entities and Attributes

1. Course (Strong Entity)

Stores information about every course offered by the university.

Attributes

- Course_Name (Primary Key)
- Department
- Course_Number

Example:

Course_Name	Department	Course_Number
DBMS	CSE	CS301

2. Section (Weak Entity)

A course can have multiple sections.

A section cannot exist without a Course, therefore it is called a Weak Entity.

Attributes

- Section_Number (Partial Key)
- Enrollment

Example:

Section_Number	Enrollment
A	60
B	55

3. Room (Strong Entity)

Stores classroom details where examinations are conducted.

Attributes

- Room_Number (Primary Key)
- Capacity
- Building

Example:

Room_Number	Capacity	Building
A101	70	Main Block

4. Exam

Stores only the examination timing because the course, section, and room details are already stored in separate entities.

Attribute

- Time

Example:

Time

10:00 AM

Step 2: Relationships

1. Course Has Section

- One course can have many sections.
- Each section belongs to only one course.

Relationship: One-to-Many (1:M)

Course 1 ----- M Section

2. Section Scheduled In Room

- Many sections may use the same room at different times.
- One room can host many section exams.

Relationship: Many-to-One (M:1)

Section M ----- 1 Room

3. Section Has Exam

Each section has one final exam.

Section 1 ----- 1 Exam

